Single-Image Depth Estimation

Results
November 2023

Alberto Rota

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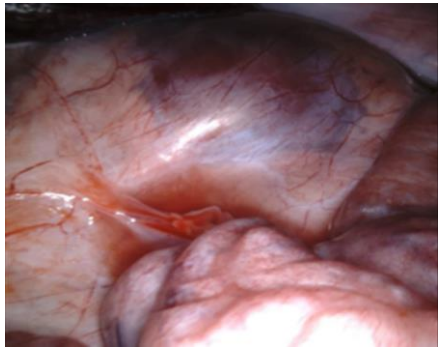
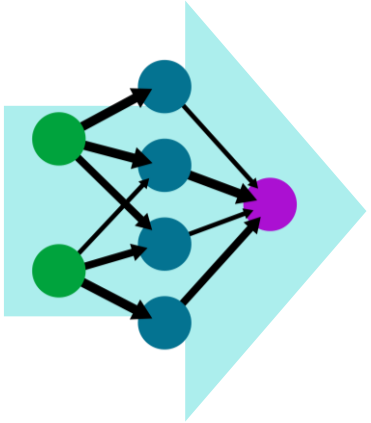
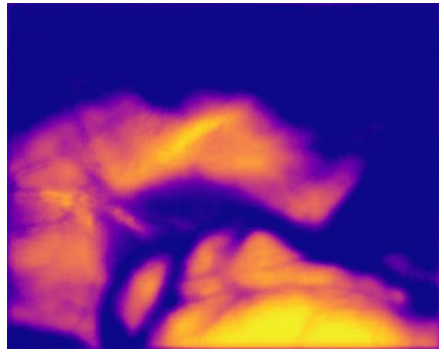
Recap: Self-Supervised Framework

From the previous frame

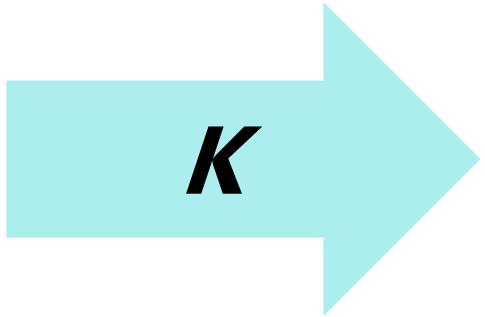




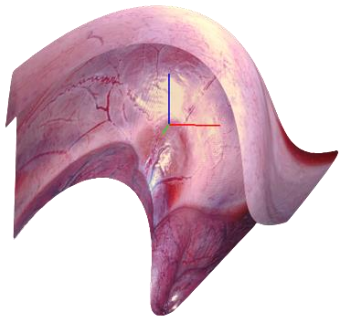
Predict the depthmap



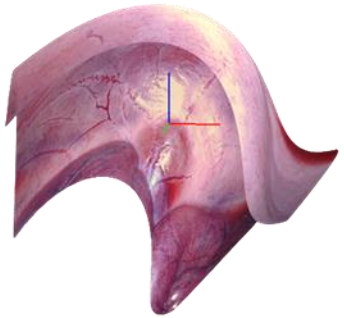
And with the Intrinsic K



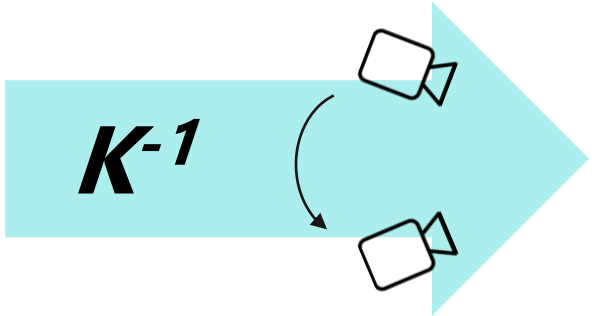
Get a RGB 3D point cloud



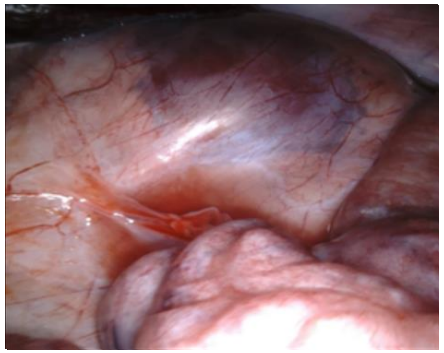
From the point cloud



The Inverse Intrinsics and the camera motion



Synthetize the current frame



The only source of error is the one made estimating the depthmap

Use this error as a loss function to Train the model

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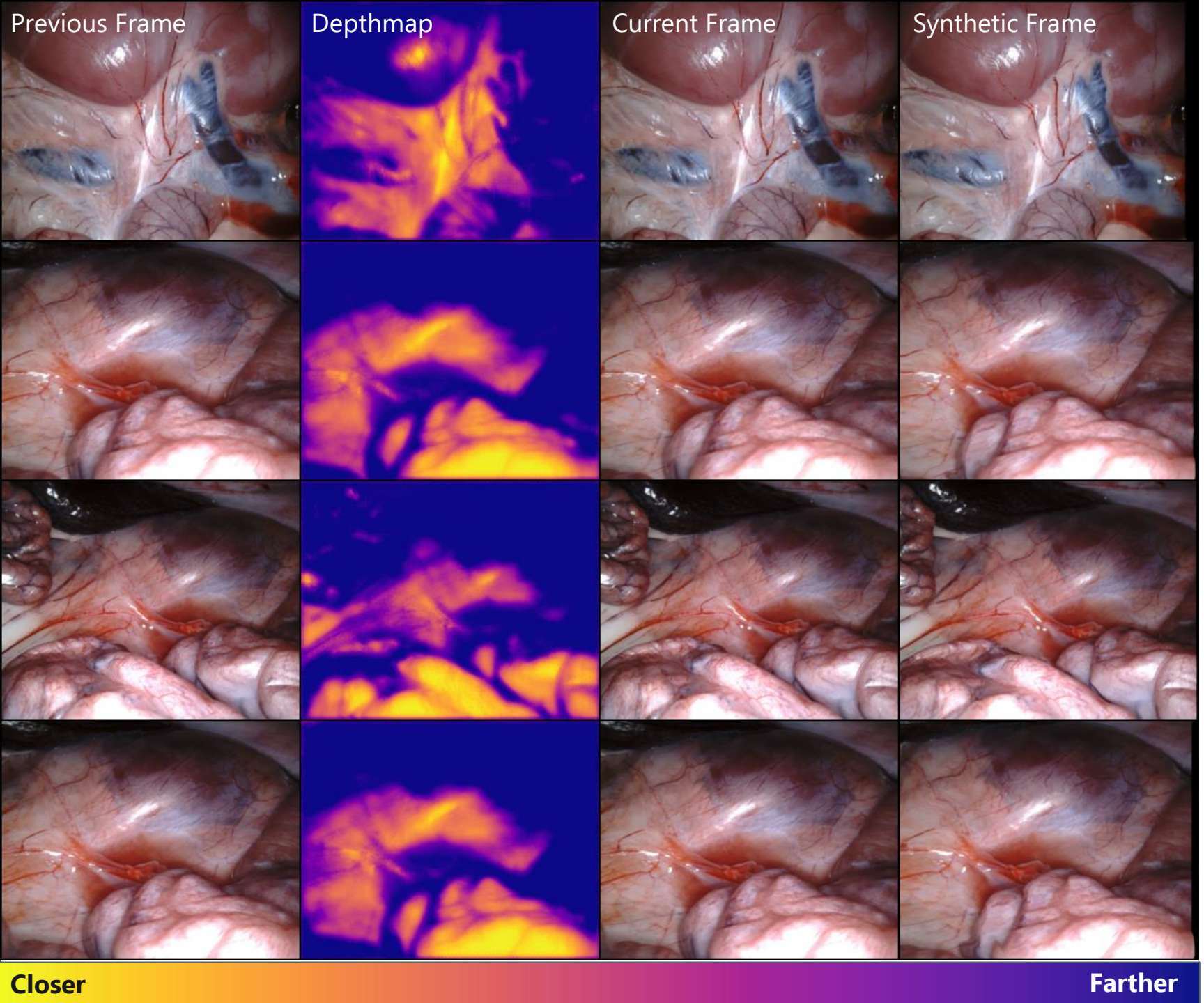

Old Results

Results from the previous Fully Convolutional model Had **~3M params**

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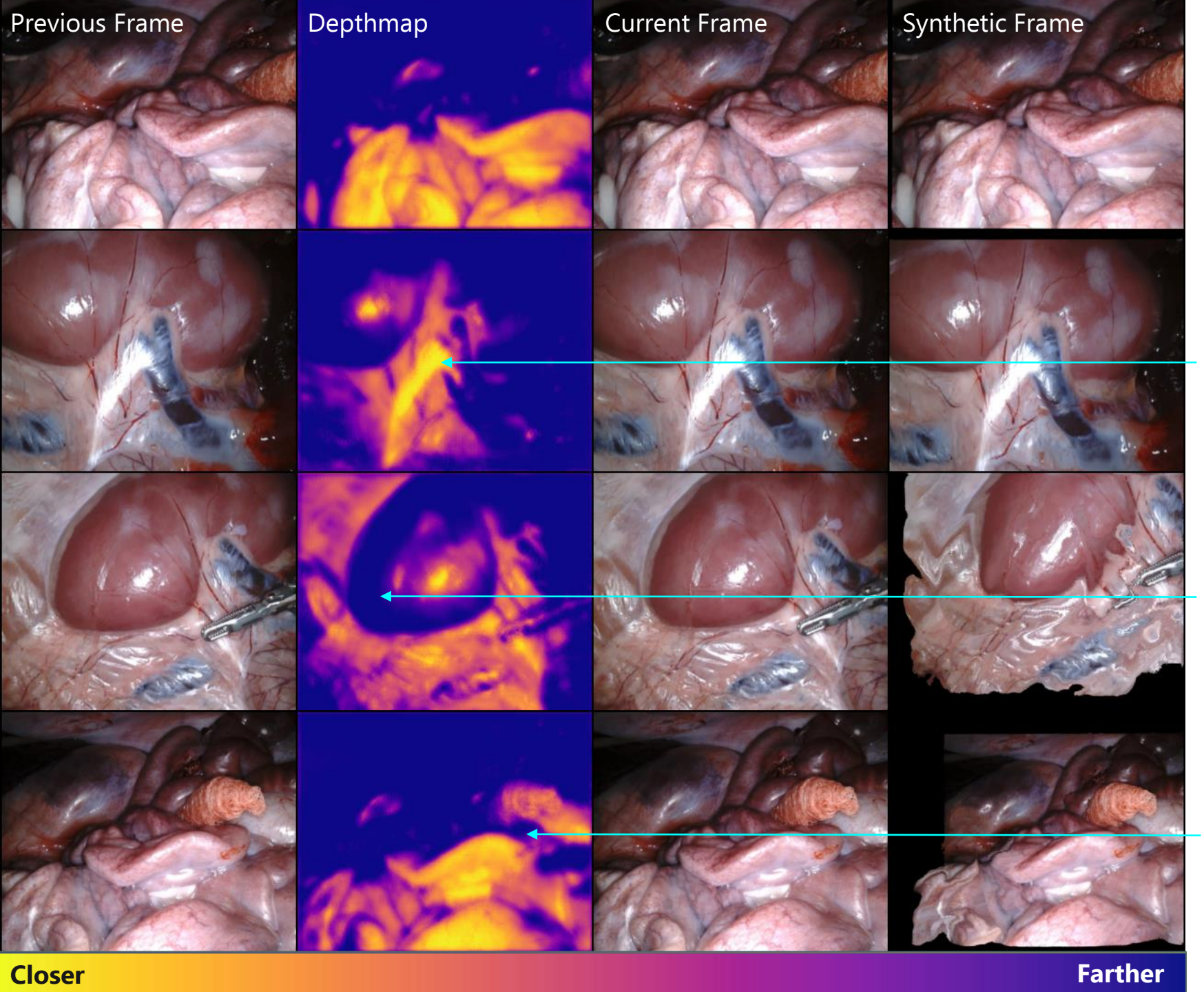
Old Results



Heavily relies on luminance

Concavity problem

More detail on regions where depth is smaller

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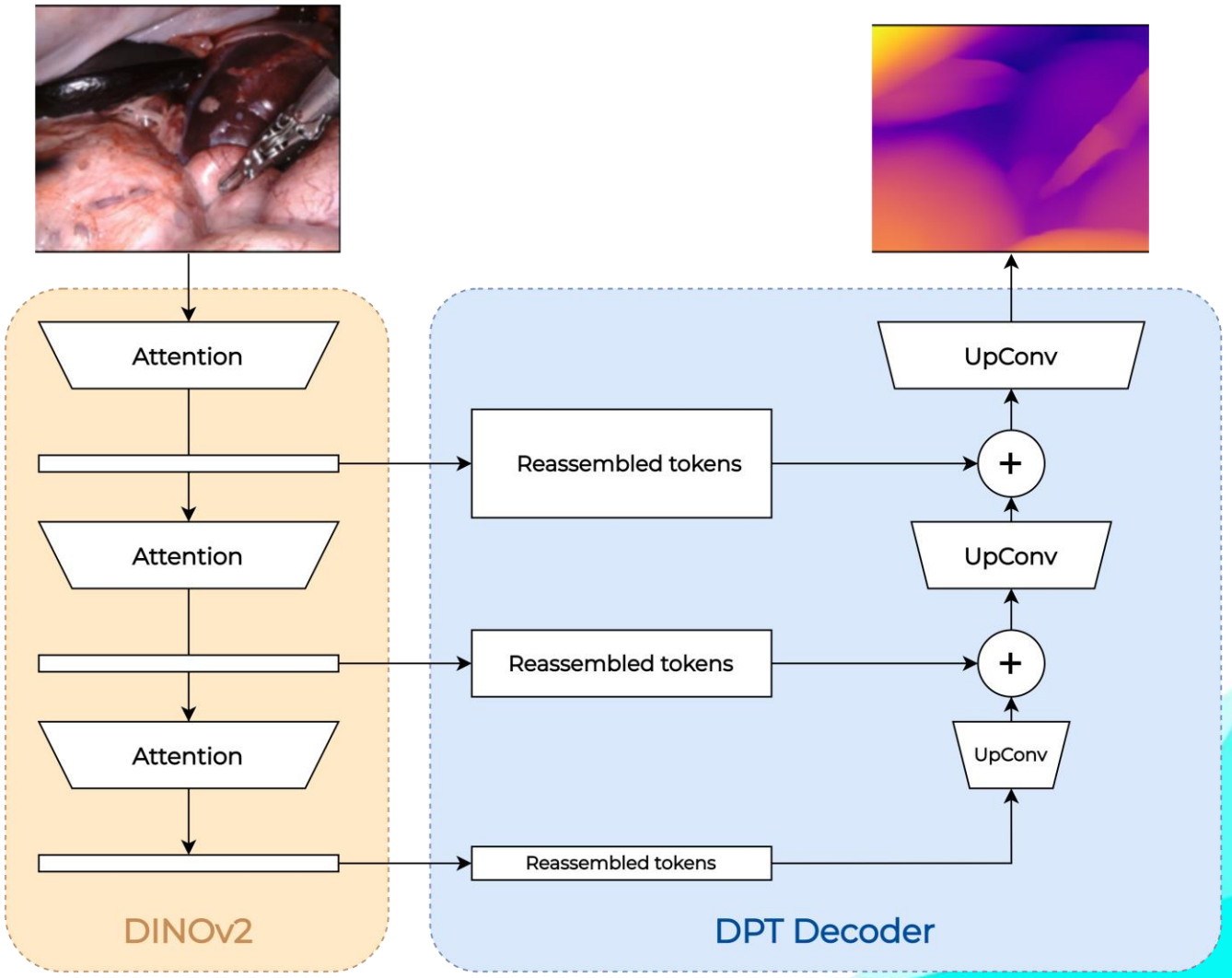
Update: Integration of DINOv2

Progress:
Introduction of **DINOv2**
as a backbone

Decoder structure from **DPT**
→ As suggested by Meta

→ Backbone frozen
→ Decoder trained from scratch on
EndoVis dataset

~300M Total Parameters
~3M Decoder parameters



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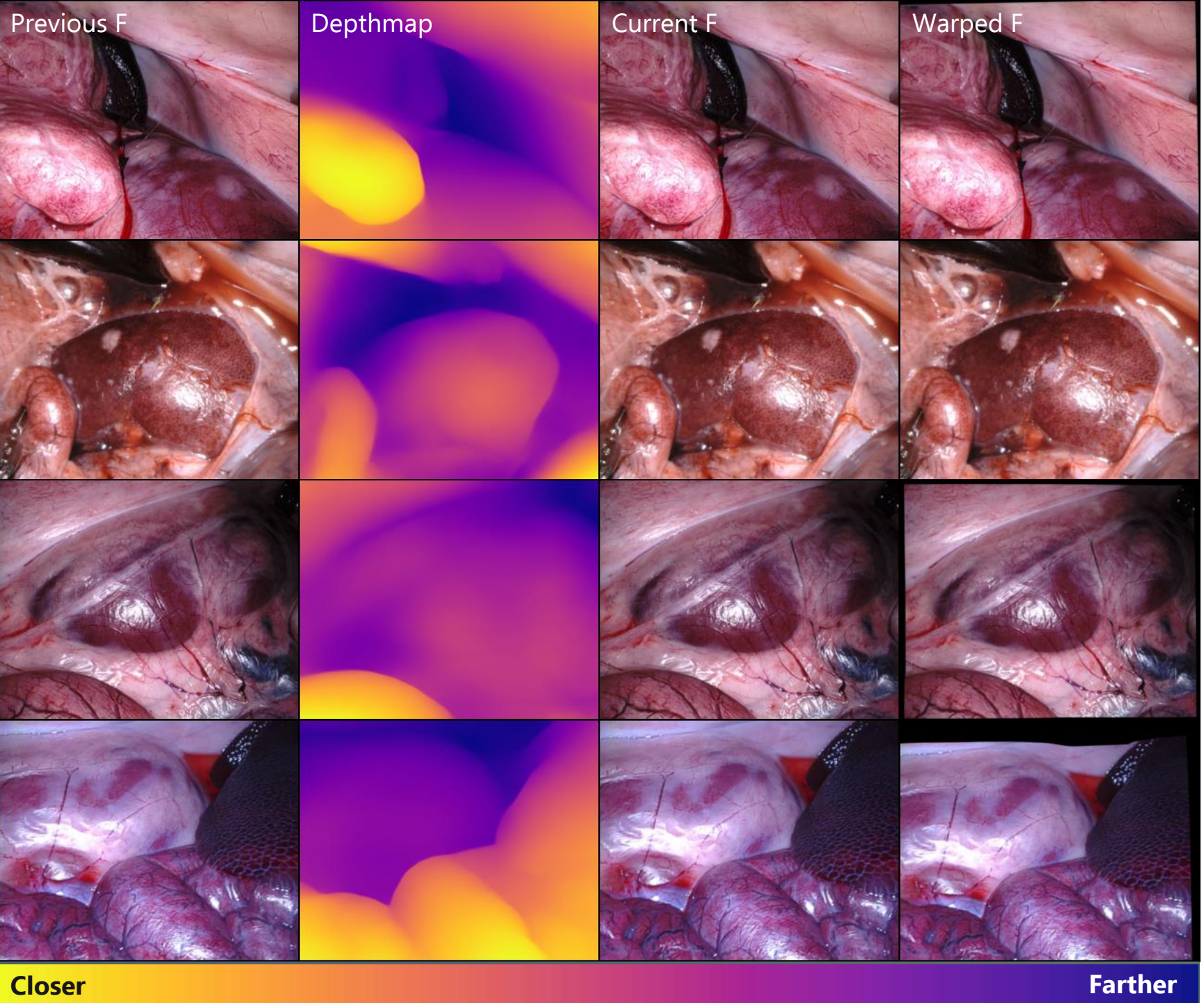
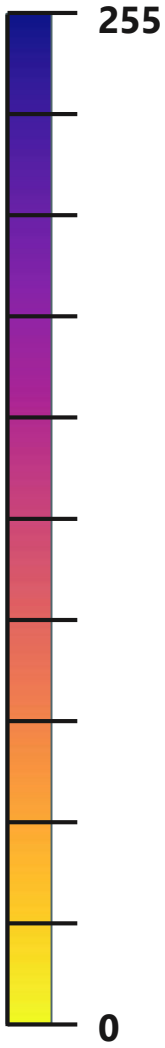
New Results





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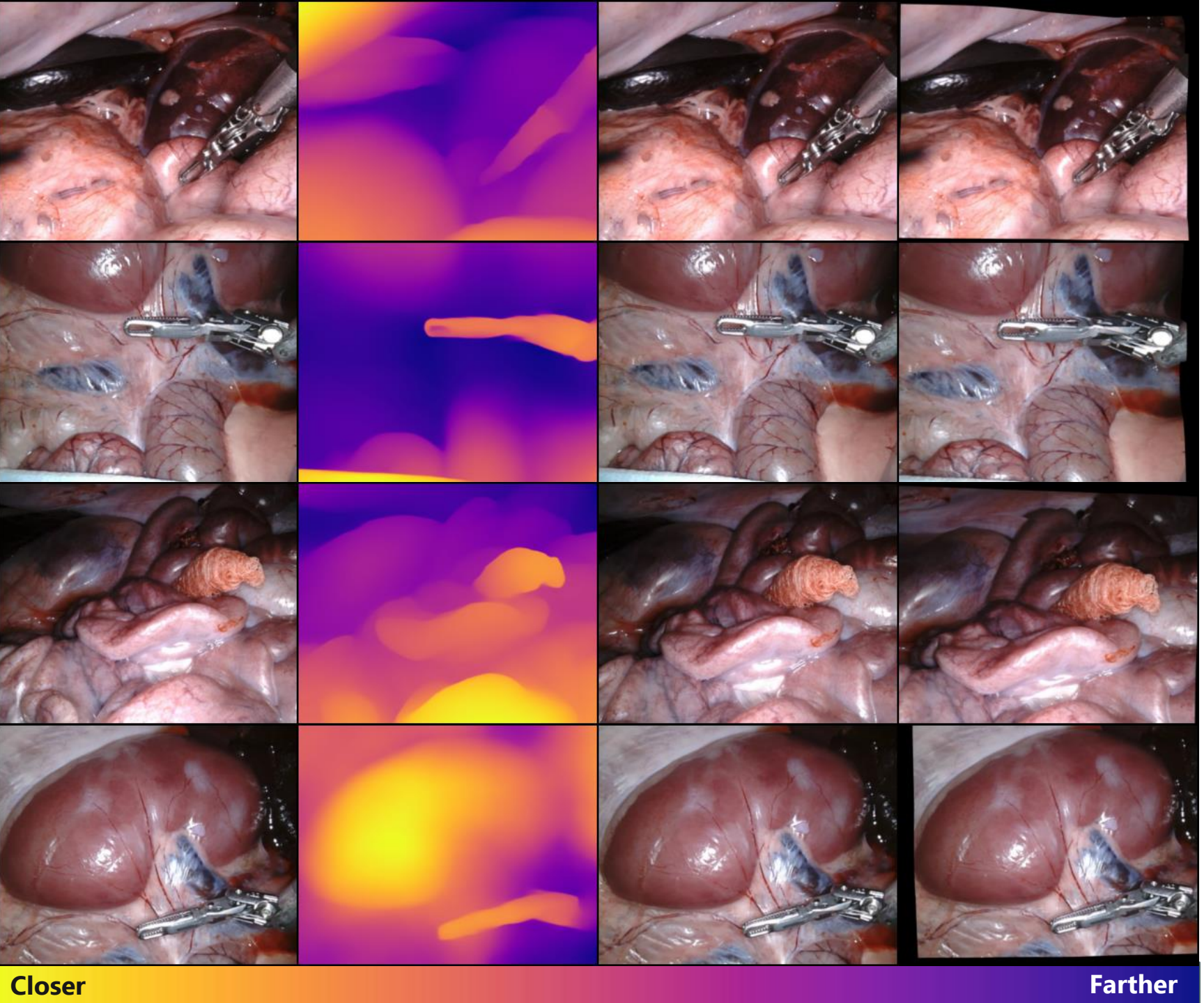
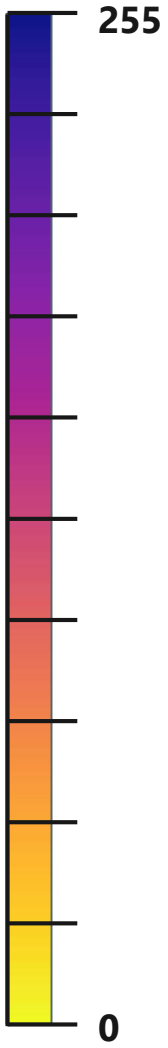
New Results





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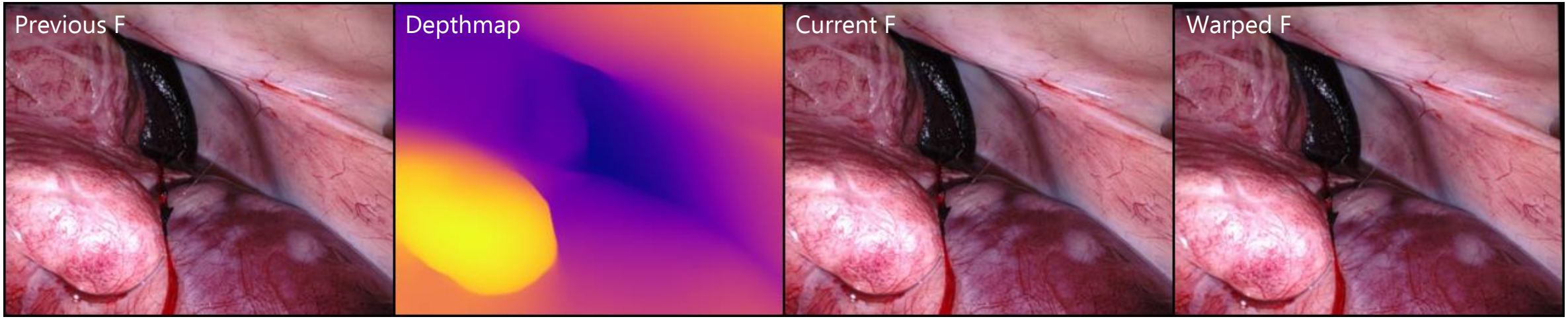
Results from DINOv2 backbone



DINOv2 + Transfer Learning to Surgical Endoscopy Videos yields
Extraordinary good results → Better than state of the art

However: No information about the real-world scale of the output depthmap →
Output sampled into 255 integer bins

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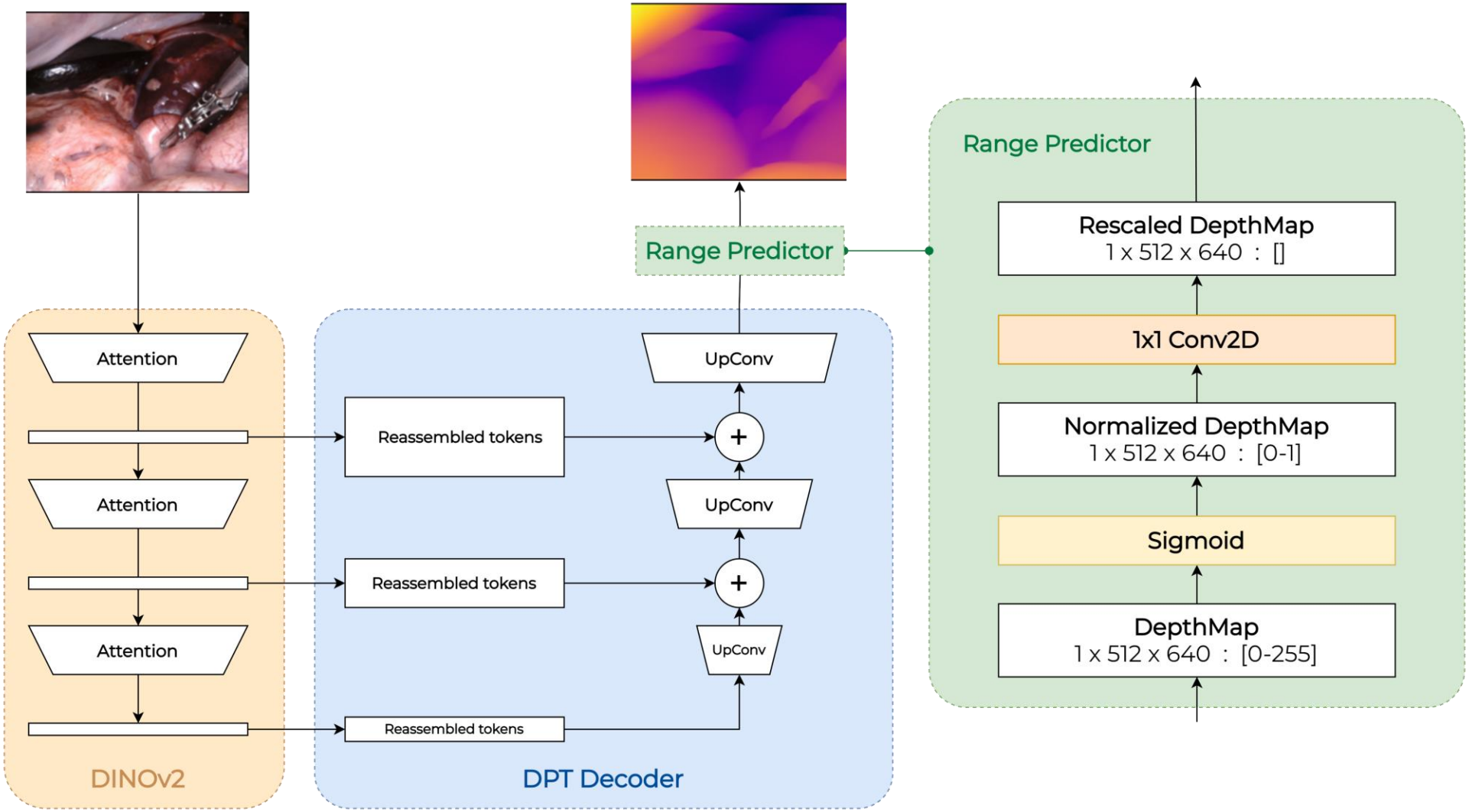
How to get the real Depth Range?



Introduction of a **learnable Range Prediction Layer**

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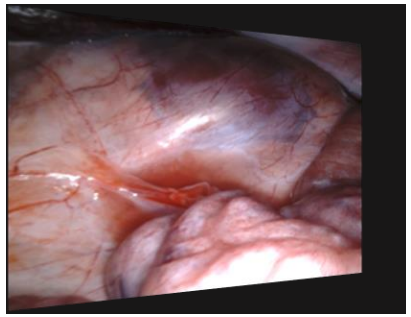
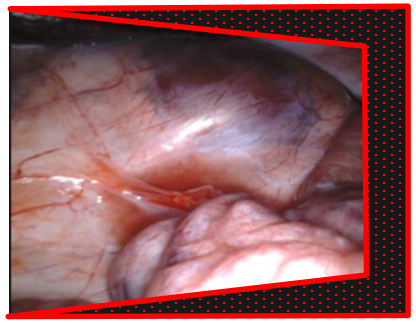
How to get the real Depth Range?

Introduction of a **learnable Range Prediction Layer**
→ **1x1 Convolution** on single channel image = **1 weight** and **1 bias**

When applied to a normalized [0-1] image, it should linearly rescale it to the real-world range → **If the weight and bias are learned optimally**

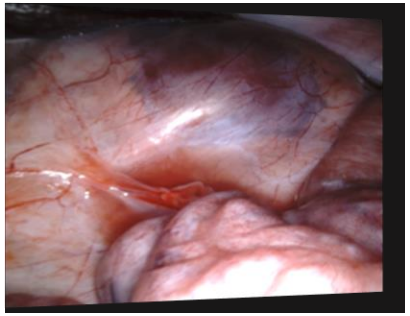
My approach: *(Not yet completely developed, still working on the technicalities)*
Optimize the Range Prediction Layer through a specific loss function that penalizes errors in depthmap range
→ **Loss**: Error on unprojected holes

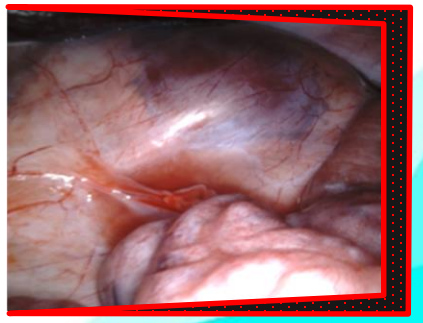




→ Big Hole

Small Hole ←





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